

EM-FDR-Bagging: Robust Estimation of the Connectivity Matrix from Multi-State Spiking Data

A bagged, scramble-calibrated wrapper around PRISM-EM

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Abstract

This document describes **EM-FDR-Bagging**, a statistical wrapper around the PRISM-EM fitter that aims to recover the shared connectivity matrix $A \in \mathbb{R}^{N \times N}$ of a non-stationary, multi-state Poisson GLM *robustly*, i.e. with explicit control of false edges and with per-edge uncertainty estimates, on real data where no ground truth exists. The method combines three ideas. First, *block bagging*: the recording is resampled into multiple overlapping bags built from spaced contiguous time blocks, and PRISM-EM is fit independently on each bag. Second, a *circular-shift null*: within each bag the fitted state sequence is frozen and the connectivity and state biases are re-estimated many times on per-neuron time-scrambled spikes, producing a null distribution of edge weights that have been passed through exactly the same estimation machinery as the real weights. Third, *per-row pooled significance with cross-bag stability selection*: edges are tested against a row-pooled null and retained only if they are selected reliably across bags. The procedure yields, for every surviving edge, a selection frequency, a bagged point estimate, a bootstrap SD, and a null-referenced effect size. We treat PRISM-EM as a black box exposed through two entry points and specify everything needed for an implementer who has the PRISM-EM code in hand.

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1 Scope and Inputs

The input is a single spike-train recording of N neurons over T discrete time bins of width Δt , in the format consumed by PRISM-EM (`spikesData/<dataName>.spikes.npz`). The target regime, which sizes every numerical choice below, is: $N = 50\text{--}500$ neurons; recordings of 10–60 minutes at $\Delta t = 10\text{ ms}$, so T ranges from 6×10^4 to 3.6×10^5 bins; per-neuron firing rates of 1–5 Hz; state-switching rates of 5–15 transitions per minute; and a small number of latent states, typically $M \approx 3$. Crucially, the method is designed for application to *real* data: it never references ground truth, and every quantity it reports is computed from the data alone.

The deliverable is not a single point estimate of A but a vetted edge list. For each off-diagonal entry (i, j) that survives the procedure we report a point estimate, a statistical uncertainty, a stability (selection) frequency, and an effect size relative to the null. Self-couplings A_{ii} are excluded from testing throughout.

2 The PRISM-EM Workhorse and Its Two Entry Points

EM-FDR-Bagging treats PRISM-EM as a callable. We summarize only what the wrapper relies on; the full model, objective, and defaults are documented separately in the PRISM-EM specification.

2.1 What PRISM-EM fits

PRISM-EM fits a non-stationary Poisson GLM in which all M latent states share one connectivity matrix A but carry state-specific bias vectors B_m . With c_t the simplex coefficients at bin t and $B_t = \sum_m c_{mt} B_m$, the rate is $\mu_t = \exp(\min(AY_{t-1} + B_t, \eta_{\text{clip}})) \Delta t$ and $Y_t \sim \text{Poisson}(\mu_t)$. Training is block coordinate descent: an E-step infers $c_{1:T}$ by simplex-projected gradient descent and Viterbi-decodes them to a hard one-hot state sequence $\hat{S}_{1:T}$; an M-step then updates $A, \mathbf{B}$ by Adam under the frozen one-hot assignment, with off-diagonal L_1 shrinkage (strength λ_3), a spectral-radius projection $\rho(A) \leq \rho_{\text{max}}$, and a learning-rate decay. The shrinkage, projection, and decay each switch on only after a configurable number of EM iterations. The fitted A is therefore the output of a specific, deterministic-in-configuration pipeline of regression plus shrinkage plus spectral projection — a fact the null construction below depends on.

2.2 Entry point 1: full fit

$$\text{full_fit}(\text{spikes}, M, \theta_{\text{EM}}) \longrightarrow (\hat{A}, \hat{S}_{1:T}, \hat{\mathbf{B}}, \text{history})$$

This is the ordinary PRISM-EM run: full E/M alternation on the supplied spikes, returning the fitted connectivity $\hat{A}$, the decoded one-hot state sequence $\hat{S}_{1:T}$, the per-state biases $\hat{\mathbf{B}}$, and the training history. Here θ_{EM} collects all PRISM-EM hyperparameters (number of EM iterations, M-step epochs, learning rates, $\lambda_2, \lambda_3, \rho_{\text{max}}$, the delay schedules, etc.).

2.3 Entry point 2: M-step-only refit with locked state sequence

$$\text{mstep_locked}(Y_{\text{prev}}, Y_{\text{curr}}, \hat{S}^{\text{lock}}, \hat{\mathbf{B}}^{\text{init}}, \theta_{\text{M}}) \longrightarrow (A, \mathbf{B})$$

This entry point runs *only* the M-step. It accepts spike pairs $(Y_{\text{prev}}, Y_{\text{curr}})$, an externally supplied and *frozen* state sequence $\hat{S}^{\text{lock}}$ (used directly as the one-hot assignment, with no E-step and no Viterbi re-decoding), and returns the re-estimated connectivity A and the refit state biases $\mathbf{B}$. The supplied $\hat{\mathbf{B}}^{\text{init}}$ is the time-ordered fit's $\hat{\mathbf{B}}$; it initializes the locked M-step but is not frozen. Re-fitting $\mathbf{B}$ in the scrambled context lets the null absorb state-specific baseline effects, so the test of A is not made artificially harsh by a bias mismatch.

Locked-state time indexing. For each lag-1 training pair (Y_{t-1}, Y_t) , the locked state used by the M-step is $\hat{S}_{t-1}^{\text{lock}}$, i.e. the state attached to the input/history bin. Equivalently, if the spike pairs have length $T - 1$, the state covariate rows are formed from $\hat{S}_{0:T-2}^{\text{lock}}$, converted to one-hot vectors. This convention is part of the model definition for lag-1 fitting and must be used identically for the real time-ordered fit and for every scrambled locked refit.

The locked M-step is configured to reproduce the real pipeline exactly: the same L_1 strength λ_3 , the same spectral projection ρ_{max} , and the same learning-rate decay as the corresponding `full_fit`. Because there is no E-step here, the EM iteration counters that gate PRISM-EM’s delayed schedules are not meaningful; instead all schedule-gated operations (shrinkage, spectral projection, decay) are **active from the first M-step epoch**. The rationale is that the null estimate must traverse the same shrinkage and constraint geometry as the converged real estimate, which lives in the late-EM regime where all schedules have already fired.

Implementation note (prerequisite). The current PRISM-EM code inlines the M-step inside the EM loop and feeds it the one-hot sequence produced internally by the E-step. To support entry point 2, the M-step must be exposed as a standalone callable that consumes an externally supplied $\hat{S}^{\text{lock}}$ and externally supplied spike pairs. This includes making the lag-1 state alignment explicit as $\hat{S}_{t-1}^{\text{lock}}$ for pair (Y_{t-1}, Y_t) . This is a small refactor of existing code rather than new modeling work; the rest of this document assumes both entry points exist with the signatures above.

3 Method Overview

EM-FDR-Bagging has two stages, and the document is organized to match the intended software structure.

Stage (a), per-bag fitting (Sections 4–6). For a single bag: assemble a resampled time series from spaced blocks, run `full_fit` to obtain the real $(\hat{A}^{(b)}, \hat{S}^{(b)})$, then run P scramble refits through `mstep.locked` with the state sequence frozen, obtaining a set of null connectivity matrices. Write everything to disk. This stage is one self-contained program; it is run N_{bag} times, once per bag, as independent jobs.

Stage (b), aggregation and pruning (Section 7). Read all per-bag outputs from a directory, build per-row pooled null distributions, perform per-bag selection and cross-bag stability selection, and emit the final vetted edge list with uncertainties.

The symbols introduced for the wrapper, kept distinct from PRISM-EM’s N, M, T , are collected in Table 1.

Symbol	Meaning	Nominal value
N_{blk}	blocks per bag	10
L_{blk}	block length (s)	60
$L_{\text{blk}}^{\text{bin}}$	block length (bins) = $L_{\text{blk}}/\Delta t$	6000
W_{src}	source window for block starts (s)	e.g. [1000, 1600]
N_{bag}	number of bags	5–10
P	scrambles per bag	4–8
$\delta_{\text{blk}}^{\text{min}}$	min separation between block starts (s)	1
$\delta_{\text{roll}}^{\text{min}}$	roll exclusion-zone half-width (s)	2
q	per-bag per-row selection quantile level	e.g. 0.99
π_{thr}	cross-bag selection-frequency threshold	0.6–0.8
M, N, T	states, neurons, bins (PRISM-EM)	≈ 3 ; 50–500; 6×10^4 – 3.6×10^5

Table 1: Symbols introduced by EM-FDR-Bagging and their nominal values. Every numeric choice in the text is written as “symbol = value” so it can be adjusted.

4 Stage (a.1): Block Bagging

A bag is a surrogate recording assembled by concatenating $N_{\text{blk}} = 10$ contiguous blocks, each of length $L_{\text{blk}} = 60\text{ s}$ (i.e. $L_{\text{blk}}^{\text{bin}} = 6000$ bins at $\Delta t = 10\text{ ms}$), drawn from a designated source window W_{src} of the original recording. The reassembled bag has length $N_{\text{blk}} \times L_{\text{blk}}^{\text{bin}}$ bins (here 6×10^4 bins, i.e. 10 minutes), which is the sequence PRISM-EM fits.

Block placement. Block start positions are bin indices drawn from the source window. The window W_{src} is given in seconds and must leave room for a full block, so admissible starts (in bins) lie in $[W_{\text{src}}^{\text{start}}/\Delta t, W_{\text{src}}^{\text{end}}/\Delta t - L_{\text{blk}}^{\text{bin}}]$. Within that range a start may fall on essentially any bin, so the number of candidate starts is large (tens of thousands), and exact collisions between two chosen starts are vanishingly unlikely. The genuine hazard is not identical starts but *near-identical, heavily overlapping* blocks: two starts one bin apart share all but one bin of content and would inject a near-duplicate stretch into the bag. We therefore impose a minimum separation $\delta_{\text{blk}}^{\text{min}} = 1\text{ s}$ (100 bins) between any two chosen starts and sample by rejection: draw N_{blk} starts uniformly from the admissible range, reject and redraw any draw that violates the minimum-separation constraint, and try up to `max_block_draw_trials=30` times for each new block. If all 30 attempts fail, accept the 31st draw regardless of the separation constraint and mark the violation in the job metadata. This bounded retry policy avoids silent infinite loops while still giving the ordinary case many chances to avoid near-duplicates.

Blocks are non-circular. Each block is a literal contiguous slice of the recording, clipped at the window end; no wrap-around is used in block extraction. Lag structure within a block is preserved exactly, so the first lag pair (Y_0, Y_1) at a block boundary is the only mild discontinuity, and with $L_{\text{blk}}^{\text{bin}} = 6000$ bins per block the boundary fraction is negligible. By default these cross-block boundary pairs are included rather than masked, because the assembled bag is passed to `full_fit` as an ordinary spike sequence and the number of such pairs is only $N_{\text{blk}} - 1$ out of $T_b - 1$. The realized block start offsets and the assembled-sequence boundary pair indices are stored in the per-bag job record so their influence can be audited, or a stricter masked-pair variant can be evaluated later.

Why overlapping bags. Bags are deliberately allowed to overlap in source time across the N_{bag} runs; this is a block bootstrap on a single finite recording. Overlap induces positive correlation between bags, which has consequences for the uncertainty estimates (Section 8) and the stability bound (Section 7); these are stated honestly where they bite. The overlap can be reduced — and its impact studied — by lengthening the total recording while holding N_{blk} small, which the parameter layout supports directly.

5 Stage (a.2): Real Fit and the Circular-Shift Null

5.1 Real fit on the bag

For bag b , call entry point 1 on the assembled bag:

$$(\hat{A}^{(b)}, \hat{S}_{1:T_b}^{(b)}, \hat{\mathbf{B}}^{(b)}) = \text{full_fit}(\text{bag}_b, M, \theta_{\text{EM}}), \quad T_b = N_{\text{blk}} L_{\text{blk}}^{\text{bin}}.$$

This $\hat{A}^{(b)}$ is the bag’s real connectivity estimate; the frozen state sequence $\hat{S}^{(b)}$ is reused unchanged for all scrambles below.

5.2 The null: per-neuron independent circular shifts

The null hypothesis for an edge is “no directed lag-1 coupling from neuron j to neuron i beyond what the bias/state structure already explains.” We realize it by destroying cross-neuron temporal alignment

while preserving each neuron’s own marginal statistics and the state timeline. Concretely, for scramble $p = 1, \dots, P$ we draw an independent integer circular shift $\delta_n^{(p)}$ for each neuron n and roll that neuron’s spike train:

$$\tilde{Y}_{t,n}^{(p)} = Y_{(t-\delta_n^{(p)}) \bmod T_b, n}.$$

Because every neuron is shifted by a different random amount, the precise lag-1 relationship that A captures is broken, while each neuron’s firing rate, burstiness, and count distribution are untouched (a circular roll is a pure permutation of a neuron’s own time axis). The state sequence $\hat{S}^{(b)}$ is *not* rolled: it is held fixed so that the null asks specifically about coupling, not about state structure.

Two-sided exclusion zone on each shift. A shift near 0 leaves the data almost unshifted and a shift near T_b wraps back to almost unshifted; both leave the real lag structure largely intact and would contaminate the null toward the alternative. We therefore restrict each shift to

$$\delta_n^{(p)} \in [\delta_{\text{roll}}^{\text{min}, \text{bin}}, T_b - \delta_{\text{roll}}^{\text{min}, \text{bin}}], \quad \delta_{\text{roll}}^{\text{min}, \text{bin}} = \delta_{\text{roll}}^{\text{min}} / \Delta t,$$

with $\delta_{\text{roll}}^{\text{min}} = 2\text{ s}$ (200 bins). The exclusion is two-sided by construction: both the small-shift and the near-full-wrap small-effective-shift regions are forbidden.

No pairwise separation among neurons. The shifts of different neurons need not be separated from one another. The null only requires that each neuron’s shift is individually non-trivial (the two-sided zone above) and that shifts are drawn independently across neurons; a coincidence in which two neurons happen to share the same shift is harmless to null validity. This matters in practice: with a 10-minute bag and a 2 s separation there would be only ~ 300 “slots,” yet a bag may contain up to 500 neurons, so any attempt to enforce pairwise distinct shifts would be infeasible by pigeonhole. We deliberately avoid that requirement. A lightweight `max_retries` guard on the per-neuron draw is retained only to handle a pathological window, not to enforce mutual spacing.

5.3 Null re-estimation through the locked M-step

For each scramble p we recompute connectivity with the state sequence frozen, via entry point 2:

$$(A^{\text{null}, (b,p)}, \mathbf{B}^{\text{null}, (b,p)}) = \text{mstep_locked}(\tilde{Y}_{\text{prev}}^{(p)}, \tilde{Y}_{\text{curr}}^{(p)}, \hat{S}^{(b)}, \hat{\mathbf{B}}^{(b)}, \theta_M).$$

The locked state attached to pair $(\tilde{Y}_{t-1}^{(p)}, \tilde{Y}_t^{(p)})$ is $\hat{S}_{t-1}^{(b)}$. The state sequence itself remains the decoded sequence from the real, time-ordered bag fit; only the spike counts entering the null M-step are circularly shifted. The biases $\mathbf{B}^{\text{null}, (b,p)}$ are initialized from $\hat{\mathbf{B}}^{(b)}$ and refit under that frozen state timeline. The configuration θ_M matches the real pipeline exactly (Section 2.3): identical λ_3 , identical ρ_{max} , identical learning-rate decay, all active from the first epoch. This is the crux of the calibration: the real $\hat{A}^{(b)}$ and the null $A^{\text{null}, (b,p)}$ are both outputs of the *same* regression, shrinkage, and spectral projection, applied to the same state timeline, differing only in whether cross-neuron alignment was destroyed and in the resulting refit biases. A significance statement of the form “this edge is larger than the same machinery produces on scrambled data” is therefore meaningful; using a raw, unshrunk M-step for the null would compare against a differently biased quantity and invalidate the test.

5.4 How many scrambles are needed

A useful efficiency follows from testing significance per row rather than per edge (Section 7). For row i we pool the $(N - 1)$ off-diagonal null weights of that row across all scrambles. A *single* scramble therefore already contributes $\sim (N - 1)$ null samples to row i ’s distribution — of order 199 for $N = 200$. With a modest $P = 4\text{--}8$ the pooled per-row null holds $(N - 1) \times P \approx 800\text{--}1600$ samples, ample to

resolve a 1% tail. By contrast a *per-edge* null would need on the order of hundreds of scrambles just to populate a single edge’s distribution, which is infeasible at this budget. The two regimes are not equivalent: the $(N - 1)$ samples within one scramble are not independent (they share a single scrambled realization of the data), so increasing P still helps by adding genuinely independent null realizations and by averaging out scramble-to-scramble variability. The recommendation is to keep P small (4–8) but greater than one, precisely because per-row pooling supplies the raw sample count while extra scrambles supply independence.

6 Stage (a.3): Per-Bag Outputs

Each per-bag job writes a self-contained file to a shared results directory, with a name carrying the bag index so that N_{bag} parallel jobs do not collide. The file contains, at minimum: the real connectivity $\hat{A}^{(b)} \in \mathbb{R}^{N \times N}$; the stack of null connectivities $\{A^{\text{null},(b,p)}\}_{p=1}^P$ of shape (P, N, N) ; the frozen state sequence $\hat{S}^{(b)}$; the realized block start indices and the assembled-sequence boundary pair indices; any block-start constraint violations caused by exhausting `max_block_draw_trials`; the realized per-neuron shifts (for reproducibility and for the data-reuse study); optionally the refit null biases $\{\mathbf{B}^{\text{null},(b,p)}\}_{p=1}^P$ for diagnostics; the fitted off-diagonal sparsity of $\hat{A}^{(b)}$ (number and fraction of surviving edges) as a sanity cross-check; and the PRISM-EM training-history summary (final objective, iteration count, whether each delayed schedule fired, final state occupancy). Stage (b) consumes only these files and never re-runs PRISM-EM.

Compute layout. Each per-bag job is a single SLURM job on a single node using 4×A100, matching the existing PRISM-EM launch (`torchrun --standalone --nnodes=1 --nproc_per_node=4`). The P scramble refits reuse the same node and the frozen $\hat{S}^{(b)}$, so they are cheap relative to the full fit. The N_{bag} jobs are mutually independent and submitted in parallel.

7 Stage (b): Aggregation, Significance, and Pruning

Stage (b) is a single postprocessing program pointed at the results directory. It performs per-row null pooling, per-bag selection, and cross-bag stability selection, then reports the vetted edge list with uncertainties.

7.1 Per-row pooled null and per-bag selection

We test significance *per row*, exploiting the assumption that the incoming edges to a given target neuron i are exchangeable under the null: they share that target neuron’s firing statistics, state occupancy, and observation scale. This is a much milder assumption than per-edge normality. The row convention is A_{ij} = effect of source neuron j on target neuron i ; therefore this row-pooled test calibrates the incoming weights to one target neuron. The separate biological interpretation of a source neuron as broadly excitatory or inhibitory is read down a column of retained edges, across the downstream targets it controls. The per-row exchangeability assumption is what makes the small scramble budget viable (Section 5.4).

For bag b and row i , pool the off-diagonal null weights across all scrambles of that bag,

$$\mathcal{N}_i^{(b)} = \{ A_{ij}^{\text{null},(b,p)} : j \neq i, p = 1, \dots, P \}, \quad |\mathcal{N}_i^{(b)}| = (N - 1) P,$$

and form a per-row threshold as the upper- q quantile of the magnitudes,

$$\tau_i^{(b)} = \text{Quantile}_q(\{ |x| : x \in \mathcal{N}_i^{(b)} \}), \quad q = 0.99 \text{ (nominal)}.$$

An edge is *selected in bag b* if its real magnitude exceeds the row threshold,

$$\text{sel}_{ij}^{(b)} = \mathbb{K}[|\hat{A}_{ij}^{(b)}| > \tau_i^{(b)}], \quad i \neq j.$$

Equivalently, each real edge can be converted to a within-row p -value against the pooled null and the Benjamini–Hochberg procedure applied within the row; the quantile-threshold form above is the operational default. The expected number of edges selected per bag and per row is recorded as a diagnostic (it is the quantity q_Λ of the stability bound below); it is *measured*, never prescribed, so the algorithm discovers how many edges it keeps rather than being told a target sparsity.

7.2 Cross-bag stability selection

A single bag’s selection is noisy. We retain only edges that are selected reliably across bags. Define the selection frequency

$$\pi_{ij} = \frac{1}{N_{\text{bag}}} \sum_{b=1}^{N_{\text{bag}}} \text{sel}_{ij}^{(b)},$$

and keep edge (i, j) in the final network if $\pi_{ij} \geq \pi_{\text{thr}}$, with $\pi_{\text{thr}} = 0.6$ – 0.8 (nominal). This is Meinshausen–Bühlmann stability selection: the data are perturbed by resampling, the base selector is run on each perturbation, and only consistently chosen variables survive.

Two ways to set the knobs (one reported, one derived). The operational default exposes the pair (q, π_{thr}) : a per-bag per-row quantile level q and a cross-bag frequency threshold π_{thr} . A more compact, single-knob alternative collapses these using the stability-selection error bound,

$$\mathbb{E}[V] \leq \frac{1}{2\pi_{\text{thr}} - 1} \cdot \frac{q_\Lambda^2}{p_{\text{cand}}},$$

where $\mathbb{E}[V]$ is the expected number of false edges, q_Λ is the *measured* average number of edges selected per bag, and $p_{\text{cand}} = N(N - 1)$ is the number of candidate off-diagonal edges. In this mode the user fixes only a single target, `target_false_edges`, interpreted as a target upper bound on $\mathbb{E}[V]$. The per-bag quantile is set loose and merely determines q_Λ , and π_{thr} is then *derived* from the bound rather than tuned. We document this route because it reduces the user’s choice to the expected number of false retained edges, but adopt the explicit (q, π_{thr}) pair as the default for transparency. The honest caveat applies to either route: the bound assumes independent (exchangeable) bags, whereas our overlapping-block bags are positively correlated, so a derived π_{thr} and any implied false-edge control should be read as approximate and reported as such, not as a guaranteed FDR.

8 Per-Edge Uncertainty: Two Distinct Errors

The design furnishes two genuinely different per-edge “errors,” which answer different questions and must not be conflated.

8.1 Null spread and effect size

For row i , the pooled null $\mathcal{N}_i = \bigcup_b \mathcal{N}_i^{(b)}$ has a mean and standard deviation,

$$m_i^{\text{null}} = \text{mean}(\mathcal{N}_i), \quad s_i^{\text{null}} = \text{SD}(\mathcal{N}_i),$$

which characterize the *noise floor*: how large an edge weight looks purely by chance once cross-neuron alignment is destroyed and the same machinery is applied. This is not the uncertainty of the estimate; it is the spread of the null. It does, however, define a natural null-referenced effect size,

$$z_{ij} = \frac{\bar{A}_{ij} - m_i^{\text{null}}}{s_i^{\text{null}}},$$

a per-edge signal-to-noise ratio against the scramble null, where $\bar{A}_{ij}$ is the bagged point estimate defined next.

8.2 Estimation variability: the across-bag bootstrap SD

The quantity one usually means by “the statistical error of an edge” is the sampling variability of its fitted weight, and bagging estimates it directly. Each bag yields a real fitted weight $\hat{A}_{ij}^{(b)}$ on a different resampled stretch of data; their spread across bags is a block-bootstrap estimate of the estimator’s sampling distribution. We report the bagged mean and bootstrap SD,

$$\bar{A}_{ij} = \frac{1}{B_{ij}} \sum_{b \in \mathcal{B}_{ij}} \hat{A}_{ij}^{(b)}, \quad \widehat{\text{SD}}_{ij} = \sqrt{\frac{1}{B_{ij} - 1} \sum_{b \in \mathcal{B}_{ij}} (\hat{A}_{ij}^{(b)} - \bar{A}_{ij})^2},$$

where $\mathcal{B}_{ij}$ is the set of bags entering the average and $B_{ij} = |\mathcal{B}_{ij}|$.

Which bags to average over. A bag in which edge (i, j) was pruned to exactly zero by L_1 contributes a structural zero, not a noisy nonzero. Two summaries are therefore reported. The *selection-conditional* estimate averages only over bags where the edge survived ($\mathcal{B}_{ij} = \{b : \text{sel}_{ij}^{(b)} = 1\}$); it reflects the typical magnitude when the edge is detected but is biased upward in magnitude. The *all-bag* estimate averages over every bag ($\mathcal{B}_{ij} = \{1, \dots, N_{\text{bag}}\}$, structural zeros included); it is diluted toward zero in proportion to how often the edge is absent. We recommend reporting the selection-conditional mean and bootstrap SD *together with* the selection frequency π_{ij} , so that the dilution is visible rather than hidden: π_{ij} and $\bar{A}_{ij}$ together convey both how often and how strongly an edge appears.

Honest caveat on calibration. The bags are overlapping block-bootstrap samples from one finite recording, not independent datasets. Overlap makes bags more alike than independent draws, so the naive across-bag SD *underestimates* the true sampling error; with only $N_{\text{bag}} = 5\text{--}10$ bags it is additionally coarse (few degrees of freedom). Accordingly $\widehat{\text{SD}}_{ij}$ should be used as a *relative* reliability measure — excellent for ranking edges and for flagging unstable ones — rather than as a calibrated frequentist standard error supporting a strict confidence interval. The degree of underestimation can be probed empirically by lengthening the recording and shrinking the block overlap (Section 4), which is a deliberate affordance of the design.

8.3 Reported per-edge record

For every edge that passes stability selection ($\pi_{ij} \geq \pi_{\text{thr}}$) the pipeline emits

$$(i, j, \pi_{ij}, \bar{A}_{ij}, \widehat{\text{SD}}_{ij}, z_{ij}),$$

together with the all-bag mean for comparison and the row null summary $(m_i^{\text{null}}, s_i^{\text{null}})$. The selection frequency π_{ij} , the bagged point estimate, the bootstrap SD, and the null-referenced effect size are complementary: an edge that is frequently selected, large relative to its bootstrap SD, and far into the null tail is trustworthy; an edge strong in one bag but rarely selected, or with bootstrap SD comparable to its magnitude, is flagged as unreliable.

9 End-to-End Summary

Stage (a), run N_{bag} times in parallel (one node, $4 \times \text{A100}$ each).

1. Assemble bag b : draw $N_{\text{blk}} = 10$ block starts in the source window W_{src} , enforce $\delta_{\text{blk}}^{\text{min}} = 1$ s separation by rejection sampling (up to `max_block_draw_trials`=30 attempts per new block, then accept the next draw and record any violation), extract non-circular blocks of $L_{\text{blk}} = 60$ s, concatenate, and record the block offsets and boundary pair indices.

2. `full_fit` $\rightarrow (\hat{A}^{(b)}, \hat{S}^{(b)}, \hat{\mathbf{B}}^{(b)})$.
3. For $p = 1, \dots, P$: draw independent per-neuron circular shifts in the two-sided zone $[\delta_{\text{roll}}^{\text{min}, \text{bin}}, T_b - \delta_{\text{roll}}^{\text{min}, \text{bin}}]$, roll spikes, and `mstep_locked` $\rightarrow (A^{\text{null}, (b, p)}, \mathbf{B}^{\text{null}, (b, p)})$ with the real pipeline's $\lambda_3, \rho_{\text{max}}$, decay active from epoch one, $\hat{S}^{(b)}$ frozen and aligned as $\hat{S}_{t-1}^{(b)}$ for pair (Y_{t-1}, Y_t) , and $\mathbf{B}$ initialized from $\hat{\mathbf{B}}^{(b)}$ but refit.
4. Write $\hat{A}^{(b)}, \{A^{\text{null}, (b, p)}\}, \hat{S}^{(b)}$, realized block starts, boundary pair indices, block-start retry violations, realized shifts, fitted sparsity, and history summary, with a bag-indexed filename.

Stage (b), single postprocessing job.

1. Per bag and row, pool the $(N-1)P$ off-diagonal null weights, form the upper- q magnitude threshold $\tau_i^{(b)}$, and select edges with $|\hat{A}_{ij}^{(b)}| > \tau_i^{(b)}$; record q_Λ .
2. Compute selection frequencies π_{ij} and keep edges with $\pi_{ij} \geq \pi_{\text{thr}}$ (or use the `target_false_edges`-derived π_{thr}).
3. For surviving edges, compute the bagged mean $\bar{A}_{ij}$, bootstrap SD $\widehat{\text{SD}}_{ij}$ (selection-conditional and all-bag), and effect size z_{ij} against the pooled row null.
4. Emit the vetted edge list $(i, j, \pi_{ij}, \bar{A}_{ij}, \widehat{\text{SD}}_{ij}, z_{ij})$ plus diagnostics.

Closing note. The method's robustness comes from never trusting a single fit: connectivity is asserted only when it survives data resampling (bagging), exceeds a null built by the identical estimation machinery (scramble calibration), and recurs across bags (stability selection). Its two reported uncertainties — the null spread and the across-bag bootstrap SD — are deliberately kept distinct, and their known limitations under overlapping bags are stated rather than papered over, so that the resulting edge list can be read with appropriate confidence on real data where no ground truth is available to check it.